

1.2.0 Let $a, b, c, d \in \mathbf{R}$ and consider each of the following statements. Decide which are true and which are false. Prove the true ones and give counterexamples to the false ones.

- a) If $a < b$ and $c < d < 0$, then $ac > bd$.
- b) If $a \leq b$ and $c > 1$, then $|a + c| \leq |b + c|$.
- c) If $a \leq b$ and $b \leq a + c$, then $|a - b| \leq c$.
- d) If $a < b - \varepsilon$ for all $\varepsilon > 0$, then $a < 0$.

(a) Let $a = 1, b = 2, c = -100, d = -1$.

$$\text{Thus } ac = 1 \cdot -100 = -100$$

$$\text{and } bd = 2 \cdot -1 = -2$$

$$\text{Then } ac < bd.$$

Hence it is false.

(c) If $a \leq b$ and $b \leq a + c$, then $|a - b| \leq c$.

pf Since $a \leq b$ and $b \leq a + c$, we obtain $b - a \geq 0$ and $b - a \leq c$

$$\text{Thus } |a - b| = b - a \quad c \geq b - a \geq 0$$

$$\text{Hence } c \geq b - a = |a - b|$$

$$\Rightarrow c \geq |a - b|$$

It is true.

1.2.3. This exercise is used in Section 6.3. The *positive part* of an $a \in \mathbf{R}$ is defined by

$$a^+ := \frac{|a| + a}{2}$$

and the *negative part* by

$$a^- := \frac{|a| - a}{2}.$$

- a) Prove that $a = a^+ - a^-$ and $|a| = a^+ + a^-$.
- b) Prove that

$$a^+ = \begin{cases} a & a \geq 0 \\ 0 & a \leq 0 \end{cases} \quad \text{and} \quad a^- = \begin{cases} 0 & a \geq 0 \\ -a & a \leq 0. \end{cases}$$

1.2.5. Let $a, b \in \mathbf{R}$.

- a) Prove that if $a > 2$ and $b = 1 + \sqrt{a-1}$, then $2 < b < a$.
- b) Prove that if $2 < a < 3$ and $b = 2 + \sqrt{a-2}$, then $0 < a < b$.
- c) Prove that if $0 < a < 1$ and $b = 1 - \sqrt{1-a}$, then $0 < b < a$.
- d) Prove that if $3 < a < 5$ and $b = 2 + \sqrt{a-2}$, then $3 < b < a$.

(a) Since $a > 2$, $a-1 > 1$

$$\Rightarrow a-1 > \sqrt{a-1} > 1$$

$$\Rightarrow a > 1 + \sqrt{a-1} > 2$$

$$\Rightarrow a > b > 2$$

Hence $2 < b < a$.

(b) If $2 < a < 3$ and $b = 2 + \sqrt{a-2}$, then $0 < a < b$.

Since $2 < a < 3$, $0 < a-2 < 1$.

$$\text{Thus } \sqrt{a-2} > a-2 > 0$$

$$\Rightarrow 2 + \sqrt{a-2} > a > 2$$

$$\Rightarrow b > a > 2 > 0$$

Hence $0 < a < b$.

(c) if $0 < a < 1$ and $b = 1 - \sqrt{1-a}$, then $0 < b < a$.

Since $0 < a < 1$, $-1 < -a < 0$,

$$\text{Then } 0 < 1-a < 1$$

$$\text{By Ex 1.3, } \sqrt{1-a} > 1-a > 0 \Rightarrow 0 > a-1 > -\sqrt{1-a}$$

$$\Rightarrow 1 > a > 1 - \sqrt{1-a} = b$$

$$\text{Ex 1.3: } x > 1 \Rightarrow x^2 > x > 1$$

$$\Rightarrow x > \sqrt{x} > 1$$

$$1 > x > 0 \Rightarrow x > x^2 > 0$$

$$\Rightarrow \sqrt{x} > x > 0$$

Hence $a > b > 0$.

$$1 - \sqrt{1-a} > 1 - \sqrt{1-0} = 1 - 1 = 0$$

(d) 跟 (b) 一樣的做法, 你們試試看。

1.2.6. The arithmetic mean of $a, b \in \mathbf{R}$ is $A(a, b) = (a+b)/2$, and the geometric mean of $a, b \in [0, \infty)$ is $G(a, b) = \sqrt{ab}$. If $0 \leq a \leq b$, prove that $a \leq G(a, b) \leq A(a, b) \leq b$. Prove that $G(a, b) = A(a, b)$ if and only if $a = b$.

① If $0 \leq a \leq b$, prove $a \leq G(a, b) \leq A(a, b) \leq b$.

Since $0 \leq a \leq b$ and $(\sqrt{a} - \sqrt{b})^2 = a + b - 2\sqrt{ab} \geq 0$,
 $a + b \geq 2\sqrt{ab} \Rightarrow \frac{a+b}{2} \geq \sqrt{ab}$

$$\Rightarrow \underline{A(a, b) \geq G(a, b)}$$

$$G(a, b) = \sqrt{ab} \geq \sqrt{a \cdot a} = \sqrt{a^2} = a$$

$$\Rightarrow \underline{G(a, b) \geq a}$$

$$A(a, b) = \frac{a+b}{2} \leq \frac{b+b}{2} = b$$

$$\Rightarrow \underline{A(a, b) \leq b}$$

Hence $a \leq G(a, b) \leq A(a, b) \leq b$.

② $G(a, b) = A(a, b) \Leftrightarrow a = b$

$$\sqrt{ab} = \frac{a+b}{2} \Leftrightarrow 2\sqrt{ab} = a+b$$

$$\Leftrightarrow 4ab = a^2 + 2ab + b^2$$

$$\Leftrightarrow \overbrace{a^2 - 2ab + b^2} = 0$$

$$\Leftrightarrow (a-b)^2 = 0$$

$$\Leftrightarrow a-b=0$$

$$\Leftrightarrow a=b$$

1.2.10. Prove that

$$(ab + cd)^2 \leq (a^2 + c^2)(b^2 + d^2)$$

for all $a, b, c, d \in \mathbf{R}$.

pf Since $0 \leq (ad - cb)^2 = a^2d^2 - 2abcd + c^2b^2$,
thus we have $a^2d^2 + c^2b^2 \geq 2abcd$.

Adding $a^2b^2 + c^2d^2$ on both side,

$$a^2d^2 + c^2b^2 + a^2b^2 + c^2d^2 \geq 2abcd + a^2b^2 + c^2d^2$$

$$\Rightarrow (a^2 + c^2)(b^2 + d^2) \geq (ab + cd)^2$$

Hence $(ab + cd)^2 \leq (a^2 + c^2)(b^2 + d^2)$, $\forall a, b, c, d \in \mathbf{R}$. //